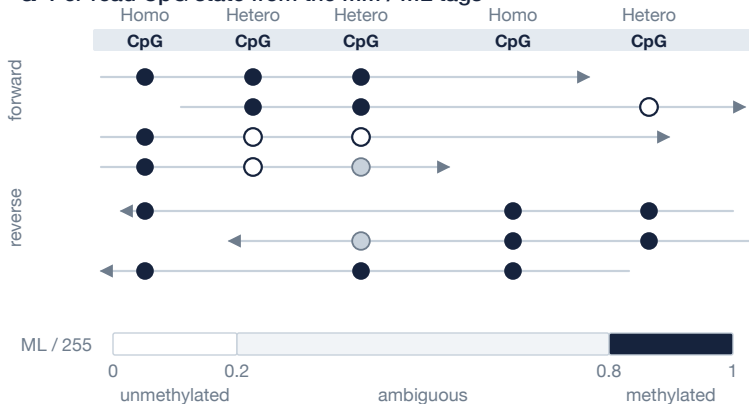
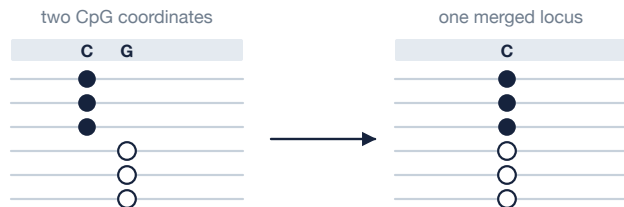


a Per-read CpG state from the MM / ML tags



b Strand pooling and merging of consecutive CpGs



5mC is called on one strand at a time; merging gives one locus with one read list

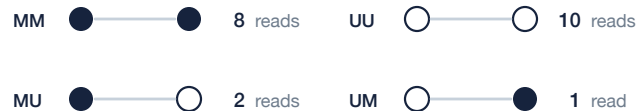
c Site genotype

heterozygous if $\min(M,U) / \max(M,U) \geq 0.6$ and $A / (M+U+A) \leq 0.2$
 otherwise homozygous methylated or unmethylated, and excluded from phasing



■ methylated □ unmethylated ■ ambiguous

d Read support between two loci



$$P = MM + UU = 18$$

$$Q = MU + UM = 3$$

$$\text{linkage} = \frac{\max(P, Q)}{P + Q} = 0.86 < 0.9 \text{ rejected}$$

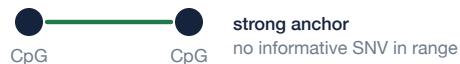
P and Q are the same quantities the phasing graph accumulates, so 5mC alleles need no methylation-specific handling in phase.

e Co-segregation with a neighbouring marker

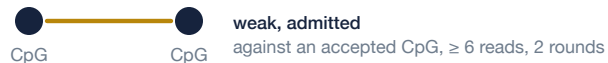
SNV-anchored



methylation-only



iterative expansion



a pair is considered only while spanning reads $> \max(6, (n_1 + n_2) / 4)$
 falling to that bound ends the neighbour scan for the site

f Output record

chr1 10469 . N . . PASS RS=P;MR=read3, read7;NR=read1, read4
 GT:MD:UD:DP 0/1:14:12:28